A short guide to the economic intuition of the asymptotic knapsack reversal dominance theorem

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Abstract. This companion document provides plain-language economic intuition for the asymptotic dominance theorems in Weinhold and Andersen (2026). Both theorems establish that in adversarial procurement under a two-value structure, a simple value-maximisation rule outperforms the textbook cost-effectiveness heuristic. The proofs use a unified fluid-limit architecture that tracks the conservation agency's actual per-turn ecological value through a deterministic ODE on the survivor pool state. Under independence, the ODE has an exact closed-form solution (certified margin +0.168); under positive correlation, it requires numerical integration (certified margin +0.214). The document explains the economic content of each proof step, the four-level self-hedging mechanism, and the role of ecological heterogeneity in generating the knapsack reversal.

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1 The Big Picture

Weinhold and Andersen (2026) show that in an adversarial land market where a conservation agency (Green) competes against agricultural developers (Farmer), a simple "buy the best ecology" rule outperforms the familiar bang-for-the-buck benchmark that underlies much conservation prioritisation.

This is surprising because the bang-for-the-buck rule (Max Efficiency) embodies the fractional-knapsack intuition that is optimal when there is no competition. The theorems show that competition reverses this logic: when your rival's purchases determine your costs, cheapness becomes a warning sign rather than an advantage.

The proof establishes this through a unified **fluid-limit architecture**: instead of bounding what Green might lose from Farmer's interference, the proof tracks what Green actually gets at each turn. As the market grows large, the per-turn ecological value converges to a deterministic path governed by an ODE on the survivor pool state. At $\rho = 0$ (independence), the ODE has an exact closed-form solution and the theorem is certified by pure algebra. At $\rho = 0.3$ (positive correlation), the ODE requires numerical integration, but the architecture is the same.

2 The Baseline Theorem ($\rho = 0$): Section by Section

2.1 The Prefix Wedge: Why Value-First Starts Ahead

Economic intuition: In the opening moves, both strategies have access to the full market. Value-first picks the ecological elite — the sites with the highest conservation value, regardless of price. The ratio rule picks sites with the best value-for-money — sites where ecological value is high RELATIVE to agricultural value.

These are different portfolios. The value-first portfolio contains the absolute best ecological sites and carries roughly the unconditional mix of agricultural values — it is not selected for

cheapness, but neither is it systematically expensive under independence. The ratio portfolio, by contrast, is explicitly biased toward low agricultural value: high E/A items are disproportionately those with anomalously low A for their E. This low-A bias makes the ratio portfolio ecologically weaker.

The prefix wedge measures the ecological quality gap between these two portfolios. At baseline, it's +0.219 per plot — a substantial lead for value-first. This gap is the engine of the entire theorem.

Why the gap exists: Under independence, the ratio E/A is high either because E is high or because A is low. A significant fraction of high-ratio items have high ratio because they are CHEAP, not because they are ecologically outstanding. The ratio rule is drawn to these items — sites that look efficient but deliver mediocre ecology.

What the formal proof does: It computes the expected ecological value of the top-mN items by E (the ME portfolio) and the top-mN items by E/A (the MX portfolio), shows the ME portfolio dominates, and quantifies the gap using population quantile integrals.

2.2 Agricultural Blindness: Why the Baseline Is So Clean

Economic intuition: Under independence ($\rho = 0$), ecological value tells you nothing about agricultural value. So when Green buys the best available ecological site, its purchase is effectively random with respect to agricultural value. Green might buy an expensive site or a cheap one — it doesn't know or care.

This "agricultural blindness" has a profound consequence for the fluid limit: Green and Farmer deplete the market at identical rates. Each turn, Farmer removes the most expensive remaining item (top of the A-distribution) and Green removes the best remaining ecology (top of the E-distribution). Under independence, neither removal is biased toward the other's dimension. The result is perfect symmetry: the Farmer frontier and Green cutoff move in lockstep.

What the formal proof does: An exact dynamic survivor law shows that, conditional on the current thresholds, the survivor pool is an iid sample from a doubly truncated host. Under independence, this host factors as a product of independent truncated normals, forcing the

frontier dynamics to be symmetric. The common trajectory satisfies $u(s) = v(s) = 1 - \sqrt{1-2s}$, where u and v are the Farmer and Green tail probabilities.

2.3 The Exact Trajectory: Closed-Form Game Dynamics

Economic intuition: Because Green is agriculturally blind, the entire game trajectory has an exact closed-form solution. Both the Farmer frontier $\alpha^*(s) = 1 - \sqrt{1-2s}$ and Green's ecological cutoff follow the same formula. The frontier accelerates — it depletes FASTER as the game progresses, because both players are removing items from a shrinking pool. Each removal is a larger fraction of what remains.

This acceleration is the mathematical expression of the "heating up" of competition as the market thins. But under independence, the heating is perfectly predictable: every game-relevant quantity (frontier position, cumulative spend, ecological value per turn) can be written as an explicit function of the game share s.

What the formal proof does: The cumulative Farmer spend is $S_F(s) = 0.1s + 3.3(1 - (1-2s)^{3/2})$. Green's cumulative spend is $S_G(s) = 0.1s + 1.65(1 - (1-2s)^{3/2})$. Green's per-turn ecological value is $e(s) = 0.1 + 9.9\sqrt{1-2s}$. All three are elementary functions of s, derived by substituting the exact trajectory into the spend and value formulas and integrating.

Why Green spends less than Farmer: Green's per-turn cost $g_A = 0.1 + 4.95\Phi(x)$ is always less than Farmer's per-turn cost $a = 0.1 + 9.9\Phi(x)$, because Green buys from the interior of the price distribution while Farmer buys from the boundary. Under independence, Green's expected purchase price is the conditional mean of A given $X \leq x$, which is the mean of the below-frontier distribution — always less than Farmer's price (the frontier value itself).

2.4 τ: When Farmer's Budget Runs Out

Economic intuition: As Farmer works down the agricultural ordering, each purchase is cheaper than the last (Farmer is buying progressively less valuable farmland). Farmer's budget drains at a rate determined by the frontier: the per-turn cost is the agricultural value at the current frontier.

Under independence, the exact Farmer spend path $S_F(s) = 0.1s + 3.3(1 - (1 - 2s)^{3/2})$ determines the exhaustion share τ by $S_F(\tau) = 2.525$. Bracketing gives $0.30471 < \tau < 0.30472$.

Green's interference slightly cheapens Farmer's menu (when Green removes a site Farmer would have bought, Farmer substitutes a cheaper alternative), which is why $\tau \approx 0.305$ exceeds the naive benchmark $\varphi \approx 0.296$. The difference (≈ 0.009) is the "interference bonus" — a quantitative expression of the self-hedging mechanism.

What the formal proof does: Evaluates S_F at two points using interval arithmetic and confirms a sign change. No ODE solver or Markov chain is needed.

2.5 Complete Allocation: $\kappa = 1 - \tau$

Economic intuition: Under budget parity, both teams start with half the total agricultural value as their budget. In the fluid limit, they collectively purchase *everything*: $\kappa = 1-\tau$ exactly. Green buys approximately $(1 - \tau)N \approx 0.695N$ items — everything Farmer didn't buy.

Why $\kappa = 1-\tau$ exactly: After Farmer exhausts at τ, Green shops alone. Green's remaining budget at τ is

$$b_G(\tau) = 2.525 - S_G(\tau) = (1 - 2\tau)(0.1 + 4.95\sqrt{1 - 2\tau}),$$

and Green's post-Farmer spend rate is the constant $0.1 + 4.95\sqrt{1 - 2\tau}$. So the continuation duration is

$$\kappa - \tau = \frac{b_G(\tau)}{0.1 + 4.95\sqrt{1 - 2\tau}} = 1 - 2\tau,$$

giving $\kappa = 1 - \tau$. Green's budget runs out at exactly the moment the last item is purchased.

(In the finite game, at most 199 items remain unclaimed — a negligible fraction of N.)

2.6 The Closed-Form Comparison: A Large Margin

Economic intuition: The fluid limit gives us Green's exact ecological value at every moment. Integrating this over the outside-prefix interval $[m, \kappa]$ gives the total ME ecological value accumulated after the prefix. Comparing this to the MX benchmark Θ_0 closes the theorem.

The margin is +0.168 per plot — larger than the simulation gap of ≈ 0.14 at moderate N. This is not an error: at finite N, the discrete game has integer effects (Farmer cannot buy exactly τN items, the last few items create boundary noise) that compress the gap. The fluid limit is the large-market limit where these frictions vanish. Conservation markets with hundreds or thousands of planning units should exhibit gaps closer to 0.17 than 0.14.

The key economic insight is unchanged from the old proof: *Value-First's continuation is good enough.* Green doesn't need to be better than MX in every period. It needs the integral of its ecological values to exceed MX's integral. The prefix lead (+0.219) is the main driver, but the fluid limit shows the continuation phase also contributes positively.

What the formal proof does: All quantities are in closed form. Combining the active-phase and post-Farmer value integrals gives the explicit outside-prefix value path:

$$V_{E,0}^{\text{out}}(\tau) = 0.1(1 - m - \tau) + 3.3(1 - 2m)^{3/2} + 1.65(1 - 2\tau)^{3/2}.$$

This function is strictly decreasing in τ (its derivative is $-0.1 - 4.95\sqrt{1 - 2\tau} < 0$), so the certified lower bound is attained at the upper endpoint of the τ-bracket. Evaluating at $\tau = 0.30472$ with interval arithmetic gives $V_{E,0}^{\text{out}} \geq 2.82336$, which exceeds $\Theta_0 = 2.6551$ by at least 0.168, certified to 80 decimal digits.

Contrast with the old proof: An earlier version of the baseline proof used worst-case deletion bounds to lower-bound Green's value in each phase separately, yielding a margin of +0.046. The fluid limit captures 3.6 times more signal by tracking what Green actually gets rather than bounding what it might lose. The old proof is retained as an alternative in an online appendix.

3 The $\rho = 0.3$ Extension: What Changes and Why

3.1 Why Correlation Amplifies the Reversal

Economic intuition: The natural expectation is that positive correlation (high ecological value tends to come with high agricultural value) should HELP the ratio rule. When the best ecological sites are expensive, a "bang for the buck" criterion seems like it should do better at finding hidden gems — sites with above-average ecology at below-average cost.

The theory shows the opposite. Under positive correlation, the high-ratio items — those with unusually high E/A — are increasingly the ones with anomalously LOW agricultural value relative to their ecology. They are the cheap-but-mediocre sites that developers aren't competing for. The ratio rule is drawn even more strongly toward this uncontested, ecologically weak corridor.

Meanwhile, the ecological elite (high E) now overlaps substantially with the agricultural elite (high A). These are the most contested sites — both Green and Farmer want them. Value-first goes straight for these contested sites; ratio-greedy avoids them precisely because they're expensive.

The prefix wedge INCREASES from +0.219 (baseline) to +0.309 ($\rho = 0.3$). Positive correlation makes the ecological quality gap between the two strategies LARGER, not smaller.

3.2 The Monotone-Argmax Lemma: Green Is No Longer Agriculturally Blind

Economic intuition: At $\rho = 0.3$, ecological value and agricultural value are positively correlated. When Green buys the best available ecological site, it's disproportionately likely to be an expensive site (high A). Green is no longer agriculturally blind — its shopping is biased toward the expensive end of the market.

This means the frontier depletion is no longer uniform. Green's picks hit the top of the A-distribution MORE often than a random draw would predict. The frontier depletes FASTER

under correlation.

The critical consequence for the proof architecture is that the baseline's exact frontier tracking is no longer available. Instead, the proof works with a lower envelope on frontier depletion (the monotone-argmax lemma below), which feeds into the cost-side certification. On the value side, the fluid-limit ODE requires numerical integration rather than closed-form algebra, but the architecture is the same: track what Green actually gets. Despite the additional machinery, the $\rho = 0.3$ margin (+0.214) is actually *larger* than the baseline margin (+0.168), because the prefix wedge increases under positive correlation.

What the formal proof does: It establishes that the truncated normal family $\mathcal{N}(\rho x, \sigma^2)$ has monotone likelihood ratio in x, which is preserved under truncation. This means: among survivors, those with higher A are stochastically more likely to have the highest E. Therefore Green's argmax-E pick is biased toward high-A items, giving a lower bound $P(\text{Green picks from top-}A\text{ block}) \geq k/M$.

3.3 The Cost Host and the Fluid-Limit Value Path

Economic intuition: Under correlation, conditioning on low A (cheap items) shifts the distribution of E downward. A pool of cheap items is, on average, ecologically poor.

For the *cost* side, this creates a tension that requires a dedicated cost host — a small, cheap, frontier-based pool $\{A \leq Q_A(\sqrt{1-2\varphi})\}$ — to bound Green's spending tightly. The maxA-topY lemma (below) operates on this host: Farmer's max-A deletions can only reduce Green's costs within it.

For the *value* side, an earlier version of the proof used a separate, larger "value host" and bounded ecological damage from Farmer's deletions via a worst-case loss kernel. That approach was too conservative: the worst-case damage overwhelmed the tiny certified margin. The current proof replaces the value host entirely with a *global fluid-limit value path* (below) that tracks Green's actual per-turn ecological value through the game dynamics. This is a fundamentally different strategy: instead of bounding what Green might lose, it tracks what Green actually gets.

This separation — cost host for spending, fluid limit for value — is unique to the correlated case. At $\rho = 0$, one pool suffices because $E \perp A$ means the ecological distribution doesn't depend on the A-conditioning.

3.4 The maxA-topY Lemma: Farmer's Self-Hedging on Costs

Economic intuition: When Farmer removes items from a pool, it always takes the most expensive one (highest A). If that item was one Green intended to buy (because it also had high E), Green substitutes a cheaper alternative — cheaper because the removed item was the single most expensive option. If Farmer's removal targets something Green didn't want, Green is unaffected.

Either way: Farmer's interference can never INCREASE Green's future spending. The self-hedging mechanism operates on costs as well as values.

This is a game-specific result, not an abstract mathematical property. It depends on Farmer buying by max-A: because the deleted item is the most expensive in the entire pool, any substitute Green needs is guaranteed to cost less. The result holds for any correlation, any distribution — it requires only that Farmer removes the current max-A item.

What the formal proof does: A two-case argument on a finite set. If the max-A item was among Green's top-v targets, Green substitutes the (v+1)-th item, which costs less (because the removed item was the global max-A). If it wasn't among Green's targets, nothing changes.

3.5 The Global Fluid-Limit Value Path: Tracking What Green Actually Gets

Economic intuition: An earlier version of the proof tried to bound the ecological damage from Farmer's deletions using a worst-case loss kernel — partitioning the ecological range into narrow bands and bounding the damage within each band. That approach failed: the worst-case cumulative damage (about 0.189 per plot) overwhelmed the tiny certified margin (0.003 per plot), even though simulations showed the actual damage was negligible.

The breakthrough came from asking a different question. Instead of bounding what Green *loses* from Farmer's interference, the proof tracks what Green *actually gets* at each turn.

At every moment in the game, the survivor pool has a definite statistical structure: it is an iid sample from a doubly truncated Gaussian copula, indexed by two moving cutoffs — the Farmer frontier x (how far Farmer has penetrated into the agricultural ordering) and the Green cutoff c (the ecological value of Green's last purchase). Green's per-turn ecological value is simply $e(c)$ — the ecological value at the current Green cutoff.

As the game progresses, both cutoffs move downward: Farmer eats into the expensive end, Green eats into the ecologically rich end. The rate at which each cutoff moves depends on both cutoffs (because Green's picks are biased toward expensive items under correlation, and Farmer's picks are ecologically non-random). This coupled evolution is captured exactly by a four-dimensional ODE system for the state (x, c, b_F, b_G) — the two cutoffs and two remaining budgets.

Why this works where worst-case failed: The worst-case loss kernel asks "what is the maximum damage Farmer could inflict?" and adds up the answer over all turns. This accumulates pessimism: each turn's bound is loose by a small amount, but the errors compound over $0.2N$ turns. The fluid limit asks "what does Green actually get, on average, at each turn?" and integrates. There is no pessimism to compound — the integral IS the actual per-turn value path, and the only approximation is the law-of-large-numbers convergence, which is standard.

The two-phase structure: The ODE has a natural switching point at Farmer exhaustion ($\tau \approx 0.313$). Before τ, both cutoffs move and both budgets drain. After τ, the Farmer frontier freezes and Green shops alone from a static pool. The switching is transversal (Farmer's budget hits zero with strictly negative derivative), so the transition is smooth.

What the formal proof does: An exact dynamic survivor law (proved by induction using order-statistics factorisations) establishes that the survivor pool is iid from the doubly truncated host $\nu_{x,c}$ conditional only on the current thresholds. This makes the state (x, c, b_F, b_G) a closed density-dependent Markov chain, to which standard Kurtz/Wormald convergence machinery applies. The ODE is initialised at $s = 0$ with thresholds at $+\infty$ and full budgets. Because the drift has a non-Lipschitz corner at $s = 0$ when $\rho > 0$, convergence is stated on $[\varepsilon, \kappa_{\text{low}}]$ for fixed $\varepsilon < m$; since the value integral begins at $m = 0.10$, nothing is lost. The ecological value integral $\int_m^{\kappa_{\text{low}}} e(c(s))\, ds$ is evaluated along this trajectory.

3.6 The Ω_φ^N Discharge: Green Can Afford to Keep Shopping

Economic intuition: The cost certificate assumes Green remains active throughout the Farmer-active phase — that Green's budget doesn't run out before Farmer's does. This needs to be proved.

The proof works by contradiction: consider the first hypothetical moment when Green's budget fails to cover the price of a remaining item. By that time, Green's cumulative Farmer-active spending is at most $c_{\text{up}} \leq 1.473N$ — the certified total of the 50 Farmer-active bands. But Green's budget is $2.525N$. The Farmer-active slack is therefore at least $1.052N$, which dwarfs any single item price (at most 10). Green can certainly afford any surviving plot, contradicting the assumption.

An important accounting point: The total certified spend in the cost certificate ($\approx 2.515N$) includes both the Farmer-active bands ($\approx 1.472N$) and the post-Farmer continuation cost ($\approx 1.043N$). The continuation cost is incurred *after* Farmer exhaustion. The Ω_φ^N discharge only concerns affordability during the Farmer-active phase, where the relevant slack is $2.525 - 1.473 \approx 1.05$, not $2.525 - 2.515 = 0.01$.

3.7 The Validated Certificate: Numerical Closure

Economic intuition: The analytical architecture reduces the theorem to a single numerical object: the value of a deterministic integral along the fluid-limit ODE trajectory. This integral is evaluated with conservative interval-Euler methods (one-sided bounds at each step), producing a certified lower bound on Green's ecological value.

The margin is large: $+0.214$ per plot at $\rho = 0.3$. This is approximately 74% of the simulation gap (≈ 0.29 per plot). The fluid-limit approach captures most of the true economic signal rather than a small residual. The remaining 26% gap between the certified margin and the simulation value reflects the conservative interval arithmetic and the fact that the proof stops at the certified affordable horizon ($\kappa_{\text{low}} = 0.621$) rather than tracking Green through full budget exhaustion.

Contrast this with the earlier worst-case architecture, which captured only about 1% of the simulation signal (margin 0.003 out of gap 0.29). The fluid limit's 70-fold improvement

in margin is not a tighter bound on the same object — it is a fundamentally different proof strategy that tracks what actually happens rather than bounding what could go wrong.

4 The Self-Hedging Mechanism: The Unifying Economic Idea

The self-hedging mechanism is the single economic idea that ties everything together. It operates on four levels:

Level 1 — Value self-hedging (the prefix wedge). When Farmer removes the most expensive items, those items are (under correlation) also ecologically valuable. Value-first has already secured the ecological elite; ratio-greedy was counting on having them available later at a good ratio. Farmer's removals systematically destroy ratio-greedy's options while leaving value-first's portfolio intact.

Level 2 — Cost self-hedging (the maxA-topY lemma). When Farmer removes the most expensive items, it inadvertently reduces Green's future spending. Each Farmer removal replaces one of Green's expensive upcoming purchases with a cheaper substitute. The rival's interference makes the value-first strategy MORE affordable, not less.

Level 3 — Value interference is absorbed by the fluid limit. Farmer's deletions do shift Green's ecological opportunities — but the fluid-limit value path accounts for this automatically. Because the proof tracks Green's *actual* per-turn ecological value (the conditional quantile from the survivor pool at each moment), Farmer's interference is already priced into the trajectory. There is no separate "damage correction" needed. The fluid limit shows that Green's ecological value per turn remains high throughout the game: Farmer and Green are still shopping in different aisles (Farmer wants expensive farmland; Green wants the best ecology), and the overlap at $\rho = 0.3$ is moderate enough that the value path stays well above the MX benchmark.

Level 4 — Budget self-hedging (the stopping bound). Under budget parity, both teams have enough money to buy nearly everything. The stopping share is determined by budget accounting, not by strategic cost management. This means the late-game dynamics (how many extra turns the ratio rule gets) are bounded by a simple counting argument rather

than a fragile cost-path calculation.

All four levels reinforce the same conclusion: the adversarial structure of the market rewards directness (buying the best ecology) and punishes cleverness (optimising the ratio), because the rival's behaviour is entangled with the focal agent's cost structure in a way that the static knapsack problem ignores.

The correlation dependence: These four levels interact differently at different correlation values, producing the hump-shaped gap observed in simulations:

- At $\rho = 0$: Level 1 is moderate (+0.219). Levels 2 and 3 are zero (Farmer's deletions are ecologically random). Level 4 provides the stopping bound. Proved margin $\geq +0.168$.
- At $\rho = 0.3$: Level 1 increases (+0.309, ratio corridor becomes more mediocre). Level 2 is active (Farmer removes expensive items, reducing Green's costs). Level 3 is absorbed by the fluid limit — Farmer's interference is priced into the value trajectory automatically, and the certified margin is +0.214. Level 4 unchanged. Gap $\approx +0.29$.
- At $\rho \to 1$: Level 1 is maximal (ratio corridor collapses). But Level 3 becomes dominant: Farmer's max-A pick IS the max-E item, causing maximum ecological damage every round. Level 2 still operates but cannot offset the value loss. Gap shrinks toward zero.

The theorem at $\rho = 0.3$ works because Level 1 (prefix quality) and Level 2 (cost savings) are both substantial, while Level 3 (value interference) is fully absorbed by the fluid-limit value path. The formal proof quantifies this through a deterministic ODE integral that captures 74% of the simulation gap, leaving a margin of +0.214 per plot.

5 What the Proof Does and Does Not Show (And What it Suggests)

The theorem is proved for a specific DGP (uniform marginals, Gaussian copula) at two correlation values. It does NOT prove:

- Universal dominance for all finite instances (counterexamples exist on very small grids)
- Dominance at all correlation levels (only $\rho = 0$ and $\rho = 0.3$ are proved, though simulations suggest it holds for all $\rho > 0$)
- Dominance under strategic Farmers (the theorem uses naïve max-A; strategic Farmers are studied only in simulations)
- Dominance under ecological compression (when E-values are nearly homogeneous, the prefix wedge disappears)

The ESP (Ecological Selection Premium) diagnostic provides a practical bridge: it measures the quality gap between the value-first and ratio portfolios on any given landscape. When ESP is large, the mechanism behind the theorem is likely to be operative and the knapsack reversal becomes more plausible. On the paper's baseline grids and on the Bolivian landscape, ESP exceeds 30%, well above the range where the theoretical conditions are satisfied.

References

Weinhold, D. and Andersen, L. (2026). Adversarial dynamics and the knapsack reversal: Insights and evidence from conservation siting. London School of Economics, working paper. Available at https://github.com/dmweinhold/Latest-Version-Papers/raw/main/Adversarial_Procurement.pdf.